

Savvy: Android R&D Feasibility Report

What Savvy's MVP can do on Android, in self-use mode and parent/child mode, and how we adjust where Android sets limits.

29 September 2026 · Tested on a Google Pixel 4 (Android 13) and a Samsung Galaxy M36 (Android 16) · Checked against current Android and Google Play policy

1. Summary

	Feature	Self mode	Parent mode	One-line reason
1	NFC card unlock	Needs NFC-card test	Needs NFC-card test	Unlocking with the card's code works on both phones. Reading a physical NFC card is not tested yet.
2	QR fallback	Feasible, with limitation	Feasible, with limitation	Scanning the card's QR unlocks (tested). A photo or screenshot of the QR also works.
3	To-do restriction	Fully feasible	Fully feasible	Apps stay blocked until the task is marked done with the card, or its time runs out. Tested.
4	App selection and blocking	Needs Play approval	Needs Play approval	Blocking works from the home screen, recent apps and links. Instant blocking uses Accessibility*, which Google must approve.
5	6/24-hour uninstall delay	Near-goal alternative	Feasible, with limitation	Self: Android does not let an app block its own removal, so the commitment comes back after reinstall. Parent: removal is blocked (tested); safe mode and factory reset still remove it.
6	Parent/child mode	n/a	Needs Play approval	Linking, parent rules reaching the child and the child lock were tested on two phones. Play's parental-control rules apply.
7	Self-use mode	Feasible, with limitation	n/a	Works. Android always lets the phone's owner switch Savvy off in the phone's Settings (see 3c).
8	Focus modes (Study, Work, Sleep)	Fully feasible	Fully feasible	Preset sessions on the same blocking engine.
9	Commitment + emergency exit	Fully feasible	Fully feasible	6 h and 24 h commitments survive restart, no internet and clock changes. Emergency exits are limited and counted.
10	Basic insights	Feasible, with limitation	Feasible, with limitation	Focus time, streaks and daily screen time per app are shown. Android keeps detail for only a few days, so Savvy keeps its own history.

Green: fully feasible Amber: feasible with an adjustment, limitation or approval Red: not feasible (none) Grey: needs validation

* **Accessibility**: an Android setting the user switches on so that an app can see which app is on screen. It lets Savvy block instantly.

2. What we have implemented

- An app picker showing the phone's apps, with the selection saved.
- Two blocking methods (Accessibility, plus a slower usage-data backup) and a Savvy block screen with card, QR and emergency options.
- Card unlock: NFC reading on the block screen, a card tap that opens Savvy, and a QR scanner. A card belongs to one account; wrong cards are refused; unlock also works offline.
- Study, Work and Sleep sessions, to-do tasks, 6 h and 24 h commitments, and a limited emergency exit.
- Blocking continues after a restart, without internet and after clock changes. The commitment is restored after reinstall and log-in.
- Parent/child: link with a code; the parent picks the child's apps and focus rules; the child can't leave parent mode without the parent's PIN; Savvy can't be removed on the child's phone; the parent sees the child's status.
- Insights: focus time, streaks and daily screen time per app (35 days kept).

3. Limitations and adjustments

	Client wants	What Android allows	Our adjustment
a	Instant blocking	Needs Accessibility, which Google Play allows only with a declaration, a demo video and an in-app explanation. Android 17's optional "Advanced Protection" setting turns it off for apps like Savvy.	The backup method keeps blocking. The blocked app's opening screen may show for under a second, with no content visible.
b	6/24 h uninstall delay (Self)	Play forbids blocking removal except in parental-control apps. The only full-control mode (Device Owner, used for company phones) needs a factory reset, so it can't be used in a Play app.	"Commitment lock": after reinstall and log-in, the same commitment and blocked apps come back (tested). Removing Savvy breaks the streak.
c	Blocked apps open only with the card (Self)	In normal use, blocked apps need the card. However, Android always lets the phone's owner go into the phone's Settings and force-stop Savvy, clear its data or remove its permissions.	Turning off one permission does not unblock, because the other method keeps blocking (tested). Savvy warns the user, and blocking resumes as soon as Savvy is opened again.
d	Child cannot remove Savvy (Parent)	Allowed when a parent authorises it. Safe mode and factory reset are outside any app's control.	Removal and Savvy's settings screens are blocked on the child's phone (tested). The parent sees when protection is off.
e	Tap the card at any time	Android does not read NFC while the phone is locked. On Android 16 and later, a user can stop cards from opening apps.	Unlock the phone first. The block screen reads the card directly either way.

f	QR as the fallback	A QR can be photographed or screenshotted and still works.	QR unlock after a short wait, or a few times per week (client decision).
g	Screen-time history	Android keeps detailed usage for only a few days.	Savvy saves its own daily totals.
h	Always on	Some phones close background apps to save battery.	A setup step to keep Savvy running, plus an automatic check every 15 minutes.
i	Block every copy of an app	Android 15's "private space" and cloned apps can hide apps from other apps.	Not tested yet. If confirmed, state it as a known gap.

4. Key questions answered

Question	Answer
Best way to block apps? Does it need Accessibility?	Accessibility for instant blocking, with the usage-data method as a backup. Blocking still works without Accessibility, just more slowly.
Will it pass Google Play?	This use is allowed with a declaration and an in-app explanation. We can't guarantee approval until Google reviews the app.
Can Self mode stop uninstall for 6/24 h?	No. The closest option is the commitment lock (limitation b).
Is Parent mode stronger?	Yes. Removal and Savvy's settings are blocked on the child's phone. Only safe mode and factory reset remain.
Restart, offline, clock change, permissions removed?	Restart: blocking continues once the phone is unlocked. Offline: blocking and card unlock still work. Clock change: ignored. Permission removed: the other method keeps blocking.
Same on different phones?	Yes on the Pixel and the Samsung we tested.
NFC: in-app, background, app closed, phone locked?	Block screen open: hold the card (1 step). Savvy in the background or closed: tapping the card opens Savvy and unlocks (1 step). Locked: unlock first. NFC off: use the QR. All of this still needs a physical-card test.
How easily can the QR be copied?	Easily: a photo works. Limit QR unlocks (limitation f).
What insights can we show?	Focus time, streaks, and total and per-app daily screen time. The parent sees the child's summary once the app tells the child about it.

5. Google Play approval risks

- **Accessibility declaration**, with a demo video and an in-app explanation plus consent.
- **"Savvy is running" notification**: a background-service declaration is needed.
- **Parent mode**: a visible notification and app icon at all times, and monitoring stated in the store listing.
- **Children as users**: Play's Families rules (no ad tracking, correct age audience).
- Low risk: listing the phone's apps uses the permitted method.

6. Needs a card, a test or a client decision

- **NFC card**: read the client's actual card; choose a cheap card or a copy-proof card.
- **Test**: Android 15 private space and cloned apps.
- **Decide**: QR limit (wait or weekly count); emergency exit rule; whether the child's own card can end parent rules; what the parent sees.

7. Client-safe wording

Can commit to:

- Block chosen apps during Study, Work and Sleep sessions, to-do tasks and 6 h or 24 h commitments.
- Unlock with the Savvy card or its QR; wrong cards are refused; works offline.
- Limited, tracked emergency exit; focus time, streaks and screen-time summary.
- Parent sets the child's blocked apps and focus rules from their own phone.

Promise carefully:

- Self mode: say "**commitment lock**: your commitment returns if you reinstall", not "uninstall protection".
- Parent mode: say "**strong protection**: the child can't remove Savvy, and the parent is told if protection is off", not "cannot be removed".
- NFC: "hold your card to the phone while it is unlocked".
- QR: "a backup to the card, with limits".
- Approval: say "built to Google Play's current rules", not "approved".